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CELL STORAGE-01-Freezing and Thawing Protocol for Adherent Cell Lines

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Protocol status: Working

We use this protocol and it's working

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Protocol Integer ID: 50648



Keywords: protocol for adherent cell line, adherent cell line, freezing procedure, effect of cell preparation method, cells viability, cell preparation method, cell morphology evaluation, cells damage, cell line, thawing procedure, cells quality, freezing process, dms0 toxic effect, increasing cells viability, cell, freezing, human umbilical vein endothelial cell, thawing protocol, following cell line, human caucasian breast adenocarcinoma, cancer histology, quality test cell, like cells morphology, useful to minimizes ice crystals formation, synovial fibroblast, freezing time, endothelial morphology, synovial fibroblasts from patient, adherent growth mode

Abstract

In our institute, cell lines are stored after freezing procedure that provides for a constant temperature lowering by 1 degree/minute. This procedure is useful to minimize ice crystals formation during freezing process, this allows to reduce the cells damage and, in turn, increasing cells viability after thawing. After storage at -80°C or in liquid nitrogen (in relation to subsequent applications or the expected freezing time) cells are thawed rapidly in a water bath at 37°C and the cryopreserving-medium is immediately removed by gentle centrifugation in order to reduce as soon as possible the DMSO toxic effect.

Currently, in our institute, the freezing/thawing procedure is validated for the following cell lines:

- **MCF7**, Human Caucasian breast adenocarcinoma, epithelial-like morphology, adherent growth mode.
- **HUVEC**, Human Umbilical Vein Endothelial Cells, endothelial morphology, adherent growth mode.
- **A549**, Human Caucasian lung carcinoma, epithelial-like morphology, adherent growth mode.
- **Synovial fibroblasts**, Human synoviocytes, fibroblast-like cells morphology, adherent growth mode.

After freezing/thawing procedure, cells quality must be assessed by cell morphology evaluation (using Optical Microscopy technique) and time required to reach confluence in the T25 cm² culture flask, which must not exceed the times indicated in the **Table 1**.

After quality test cells must be resuspended according to specific protocols (for details see **Table 2**).

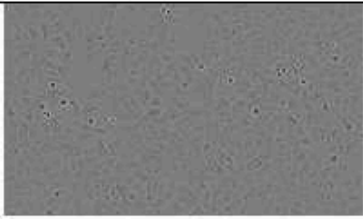
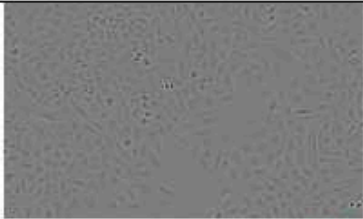
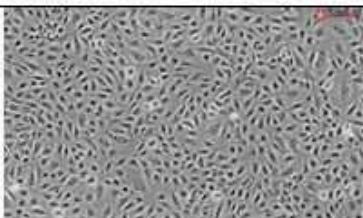
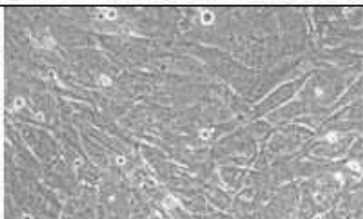
Cell lines	Time to Reach Confluence	Reference for morphology evaluation
MCF7	48 hours	
NSCLC A549	24 hours	
HUVEC	72 hours	
Synovial fibroblast	96 hours	

Table 1

Cell line	Culture medium	Applications post thawing	Our publication using this applications
NSCLC A549	RPMI 1640+10%FBS+100 U/ml penicillin/streptomycin	Cell cultures	Coelho et al., 2019
HUVEC	EndoGRO™basal medium+ 2% FBS+10 mM L-glutamine+0.75 U/ml Heparin sulfate+ 5 ng/ml rh-VEGF + 5 ng/ml rh-EGF + 5 ng/ml rh-FGF basic+ 15 ng/ml rh-IGF-1 + 50 µg/ml ascorbic acid+ 100 U/ml penicillin/streptomycin	Cell cultures	Marino et al., 2017
MCF7	DMEM without phenol red+10% FBS+100 U/ml penicillin/streptomycin+2 mM L-glutamine	Cell cultures	Spagnuolo et al., 2014; Cosentino et al., 2007
Synovial Fibroblast	RPMI 1640 without phenol red+10% FBS+100 U/ml penicillin/streptomycin	Cell cultures	Capellino et al., 2014

Table 2

List of published work by our institute that using this procedure

- Coelho M, Imperatori A, Chiaravalli AM, Franzi F, Castiglioni M, Rasini E, Luini A, Legnaro M, Marino F, Ribeiro L, Cosentino M. **Beta1- and Beta2-Adrenoceptors Expression Patterns in Human Non-small Cell Lung Cancer: Relationship with Cancer Histology.** J Neuroimmune Pharmacol. 2019 Dec;14(4):697-708. doi: 10.1007/s11481-019-09879-6. Epub 2019 Oct 16. PMID: 31620969.
- Marino F, Schembri L, Rasini E, Pinoli M, Scanzano A, Luini A, Congiu T, Cosentino M. **Characterization of human leukocyte-HUVEC adhesion: Effect of cell preparation methods.** J Immunol Methods. 2017 Apr;443:55-63. doi: 10.1016/j.jim.2017.01.013. Epub 2017 Feb 3. PMID: 28167274.
- Spagnuolo P, Rasini E, Luini A, Legnaro M, Luzzani M, Casareto E, Carreri M, Paracchini S, Marino F, Cosentino M. **Isoflavone content and estrogenic activity of different batches of red clover (*Trifolium pratense* L.) extracts: an in vitro study in MCF-7 cells.** Fitoterapia. 2014 Apr;94:62-9. doi: 10.1016/j.fitote.2014.01.027. Epub 2014 Feb 5. PMID: 24508860.
- Capellino S, Cosentino M, Luini A, Bombelli R, Lowin T, Cutolo M, Marino F, Straub RH. **Increased expression of dopamine receptors in synovial fibroblasts from patients with rheumatoid arthritis: inhibitory effects of dopamine on interleukin-8 and interleukin-6.** Arthritis Rheumatol. 2014 Oct;66(10):2685-93. doi: 10.1002/art.38746. PMID: 24965369.
- Cosentino M, Marino F, Ferrari M, Rasini E, Bombelli R, Luini A, Legnaro M, Delle Canne MG, Luzzani M, Crema F, Paracchini S, Lecchini S. **Estrogenic activity of 7-hydroxymatairesinol potassium acetate (HMR/lignan) from Norway spruce (*Picea abies*) knots and of its active metabolite enterolactone in MCF-7 cells.** Pharmacol Res. 2007 Aug;56(2):140-7. doi: 10.1016/j.phrs.2007.05.001. Epub 2007 May 22. PMID: 17572100.



Materials

Reagents

- **FBS** catalog number: ECS0180L-500 ml, Euroclone
- **RPMI 1640** catalog number: ECM0495L-500 ml, Euroclone
- **RPMI 1640 without phenol red** catalog number: ECM0505L-500 ml, Euroclone
- **DMEM** catalog number: ECM0160L-500 ml, Euroclone
- **Penicillin/streptomycin** catalog number: ECB3001D-500 ml, Euroclone
- **L-Glutamine** catalog number: ECB3000D-500 ml, Euroclone
- **EndoGRO™-VEGF complete media kit** catalog number: SCME002, Millipore
- **Trypsin/EDTA** catalog number ECB3052D, Euroclone
- **PBS** catalog number ECB4053L, Euroclone
- **DMSO** catalog number 276855, Sigma-Aldrich

Sterile plastic disposable:

- **Cryogenic vials** catalog number: CC430659, Corning
- **T25 cm² culture flasks for adherent cells** catalog number: CC430639, Corning
- Generic laboratory plastic disposable

Instrumentation required:

- Laminar flow hood
- Centrifuge
- -80°C freezer

Before start

Work under laminar flow hood when you are processing samples (from the beginning to the end of the following procedure). Make sure you are using **sterile plastic disposable and medium**.



CELL FREEZING PROCEDURE FOR ADHERENT CELLS (MCF7, HUVEC, A549, SYNOVIAL FIBROBLAST)

- 1 At confluence remove culture medium from the flasks, wash twice the cell monolayer with  5 mL of sterile PBS.
- 2 Add  1 mL (for T25 cm² culture flasks) of trypsin/EDTA for 5-10 minutes, observe the cells under a microscope to make sure that cells are detached.
- 3 Add  4 mL of complete medium to inactivate trypsin, recover the cell suspension in 15 mL conical tube and centrifuge for  00:05:00 at  200 x g  Room temperature .
- 4 Remove supernatants and resuspend the cell pellet in  1 mL of cryopreserving medium (90% FBS + 10% DMSO).
- 5 Immediately aliquot  1 mL of cell suspension into a cryogenic vial.
- 6 Place the cryogenic vial(s) in cotton wool, and then place it in a polystyrene box.
- 7 Put the polystyrene box in -80°C freezer.

Equipment

-80°C Freezer

NAME

Eppendorf

BRAND

B U9230-0001

SKU



- 8 After a minimum of  24:00:00 transfer the cryogenic vials into a box for storage in -80°C freezer.

CELL THAWING PROCEDURE

- 9 Remove cryogenic vial from -80°C and keep it at  37 °C in water bath, this procedure should take approximately 1-2 minutes or until ampoule is slightly thawed.
- 10 Clean the cryogenic vial outside with 70% EtOH, open the cryogenic vial and add  1 mL of prewarmed complete medium for specific cells lines (**Table 2**).
- 11 Transfer the cell suspension into a 15 mL conical tube containing  10 mL of prewarmed complete medium for specific cells lines.
- 12 Centrifuge for  00:05:00 at  200 x g at  Room temperature
- 13 Resuspended the cells into appropriate cell culture medium (**Table 2**) and start quality control procedure (**Table 1**).